

Project Plan:

Goals: Over the course of this project, we want to analyze how physical characteristics of a basketball player are predictors of a player's performance. We will plan to gain insights using RStudio and present our findings.

Needs: Basketball wingspan dataset

github repository

detailed plan

wrangled dataset

description of the data

description of fair and care principles

one plot and one table per group member

additional files for repository

qmd and pdf

citations document

presentation

peer and self-evaluations

Step-by-Step:

1. create a detailed plan of assignment
2. research to find a dataset
3. ensure dataset meets requirements
4. wrangling/tidy the dataset
5. find attributes to focus on
6. write code for plots and tables
7. ensure fair and care principles are met

8. provide analysis on plots and tables
9. prepare qmd file to report of analysis with visuals
10. ensure repository meets all requirements (citations)
11. present
12. finish peer and self-evaluations

Url to where we found the dataset: [NBA Wingspan & Performance \(Combined Dataset\) – SCORE Sports Data Repository](#)

Visualizations:

Pair 1: Physical Characteristics by Position (Sean)

Table: A grouped summary table showing mean and SD of height_inches, wingspan_inches, and wingspan_advantage for each of the five positions (PG, SG, SF, PF, C).

Visualization: A side-by-side box plot with position on the x-axis and wingspan/height on the y-axis, faceted to show both measurements.

Pair 2: Performance by Wingspan Advantage Tier (Charlie)

Table: Players binned into Low (<4"), Medium (4–6"), and High (>6") wingspan advantage tiers, with average blk, trb, stl, and pts for each group.

Visualization: A faceted bar chart with wingspan tier on the x-axis and separate panels for each stat (blk, trb, stl). If the hypothesis holds, you'd see a clear staircase pattern from Low to High across defensive stats.

Pair 3: Player-Level Wingspan Extremes and Performance (Suhas)

Table: The 10 players with the highest and 10 with the lowest wingspan_advantage, showing their position, height_inches, pts, blk, trb, and stl.

Visualization: A scatter plot with wingspan_advantage on the x-axis and blk or trb on the y-axis, points colored by position, with player names labeled at the extremes using ggrepel. Shows the continuous relationship and whether it holds across positions.

Items to submit:

README.md

QMD file

pdf render of QMD file

Project plan (this)